CONFIDENTIAL

SCHOOL OF COMPUTING
Faculty of Engineering

UNIVERSITI TEKNOLOGI MALAYSIA

SKILL-BASED TEST 1
SET C

SEMESTER II 2019/2020

SUBJECT CODE	**: SECJ/SCSJ 1023**
SUBJECT NAME	**: PROGRAMMING TECHNIQUE II**
YEAR/COURSE	**: 1 (SECB/SECJ /SECP/SECR/SECV)** **2 (SCSR/SCSV)**
TIME	**: 2.45 pm – 4.45 pm (2 Hours)**
DATE/ DAY	**: 30th APRIL 2020 (THURSDAY)**

INSTRUCTIONS TO THE STUDENTS:

- Please read the ***General Guidelines for the Programming Technique II Test*** that is shared in WhatsApp's Group and/ or provided in UTM e-learning
- Read the problem and instructions carefully.
- You are given **TWO HOURS** to complete the test inclusive the submission of your program.

IMPORTANT NOTES:

- All the **COMMENT STATEMENTS** in the submitted program **WILL NOT BE EVALUATED**.

SUBMISSION PROCEDURE:

- Only the source code is required for the submission and the source code's file shall be named as follows: ***Name_ matricsNo_section.cpp*** (i.e. *AinaAli_A19EC018_01.cpp*).
- You do not need to compress the file.
- Submit the source code file (i.e., sbt1.cpp) via the **UTM's e-learning system**.

Problem [100 Marks]

Write a complete C++ program that contains a class called **Student**.

1. The class should have seven (7) **`private`** member variables (attributes) to hold a name, matric number that has nine characters, Grade Point Average (GPA), Cumulative Grade Point Average (CGPA), credit earned this semester, cumulative credit obtained for all semesters, and conferment result.

2. The class should have the following **`public`** member functions:

   - A **constructor with default arguments** that assign the values of name, matric number, CGPA and current total credit earned member variables with the passed arguments. The constructor also enables to assign empty string ("") to the name and matric member variables; and zero (0) to the GPA, CGPA, current total credit earned, and total credit earned for all semester member variables. It is also need to invoke **updateCGPA** function to calculate the latest CGPA.

   - A **getName** function that returns the value of name member variable.

   - A **getStatus** function that returns the status of conferment result member variable. The conferment result is based on the following table:

CGPA	Conferment Result
Below 1.7	Fail Status
$1.7 \leq \text{CGPA} < 2.0$	Probation Status
2.0 and above	Good Status

   - A **updateCGPA** function that update the student CGPA based on the GPA student for that semester. The calculation of the new value of CGPA member variable is made according to the following formula:

$$\mathbf{CGPA_{new}} = \frac{\mathbf{(GPA \times CE) + (CGPA_{current} \times CCO)}}{\mathbf{CE + CCO}}$$

   where CE is current total credit earned, CCO is total credit earned for all semester and $\text{CGPA}_{\text{current}}$ is current CGPA.

   - A **readInput** function that reads the values from the keyboard for name, matric number, GPA, CGPA, current total credit earned, and total credit earned for all semester member variables. It is also need to invoke **updateCGPA** function to obtain the new value of CGPA.

- An **<u>overloaded equality (==) operator</u>** that determines if two objects have the same status of conferment result, then return **true**; otherwise return **false**.

3. The class should also have the following **`friend`** of standalone functions:
   - A **<u>dispInfo</u>** function to display the year and program of study of Student's object in the following format (please refer to the sample output given in **Figure 1)**:

     ```
     You are in year # in ????? ?????
     ```

     The year of study will be determined by the second and third characters of matric number, whereas program of study will be determined by the fourth and fifth characters of matric number based on the following table:

Abbreviation	Program
CS	Computer Science
EC	Engineering Computing
Apart from CS and EC	Non Computing Program

     Example: *Matric number is A16CS0233. The second and third characters are 16, meaning the Student's object is fourth year student. The fourth and fifth characters are CS. Therefore, the Student's object is study in Computer Science Program.*

     Note: You must use an appropriate function(s) in **`string`** class to determine the year and program of study of Student's object.

   - An **<u>overloaded output (<<) operator</u>** function to display the object in the following format (please refer to the sample output given in **Figure 1**):

     ```
     Hello, ????? ?????
     You are in year # in ?????
     Your matric number is ?????
     Your current CGPA is #.##
     Your status is ?????
     ```

     Note: If the current CGPA of Student's object is 3.67 and above, then display the following message:

     ```
     CONGRATULATIONS. You are awarded DEANs LIST
     ```

4. Write an appropriate **<u>main</u>** <u>function</u> that do the following tasks:
   - Create two **Person** objects named **p1** and **p2**. The objects have the following data:

Object	p1	p2
Name	""	`Hashim Abu`
Matric number	""	`A17CS1234`
CGPA	`0.0`	`4.0`
Cumulative credit obtained for all semesters	`0`	`18`

- Using an appropriate function(s), read the required input values and print the data of the appropriate object along with the notes about status of conferment result for two objects (**p1** and **p2**) either both have same or different status.

Figure 1 shows a sample screen output that your program should produce. Note that, **bold** texts indicate keyboard input entered by the user. **Table 1** is the assessment criteria for the program.

Table 1: Assessment criteria

Item	Criteria	Marks
A	i) The program is able to run.	5
	ii) Using an appropriate structure for the program	
	- the code is properly indented	1
	- all the required header files are included	1
	- the main function is properly written including proper input and output formatting used	2
B	The definition of class Person	1
	i) Data members	5
	ii) Constructor	10
	iii) getName function	2
	iv) getStatus function	8
	v) updateCGPA function	8
	vi) readInput function	9
	vii) Overloaded operator==	7
C	The definition of friend functions	
	i) dispInfo function	17
	ii) Overloaded output (<<) operator	11
D	Function main	
	i) Object p1 and p2 creation	3
	ii) Read the required input values	1
	iii) Display the data of appropriate object	3
	iv) Display the notes about weight status for two objects	6
Total		**100**

```
CGPA Calculator

<<< Input >>>
Name: Abdullah Sani
Matric Number: A16CS0233
CGPA: 3.4
Total Credit: 87
GPA: 3.56
Credit earned this semester: 16

<<< Output >>>
Hello, Abdullah Sani
You are in year 4 in Computer Science
Your matric number is A16CS0233
Your current CGPA is 3.42
Your status is Good

Hello, Hashim Abu
You are in year 3 in Computer Science
Your matric number is A17CS1234
Your current CGPA is 4.00
Your status is Good
CONGRATULATIONS. You are awarded DEANs LIST

**Note: Abdullah Sani and Hashim Abu have Good status.
```

(a) Example of run 1

```
CGPA Calculator

<<< Input >>>
Name: Salina Hamid
Matric Number: A18EC0217
CGPA: 2.01
Total Credit: 51
GPA: 1.96
Credit earned this semester: 18

<<< Output >>>
Hello, Salina Hamid
You are in year 2 in Engineering Computing
Your matric number is A18EC0217
Your current CGPA is 2.00
Your status is Probation

Hello, Hashim Abu
You are in year 3 in Computer Science
Your matric number is A17CS1234
Your current CGPA is 4.00
Your status is Good
CONGRATULATIONS. You are awarded DEANs LIST

**Note: Salina Hamid and Hashim Abu have different status.
```

(b) Example of run 2

Figure 1: Example runs of program